# Signal Monitoring TO BE 0.7.6.4.19.1

Review design for a planned Radar search pipeline change

Status: Implemented (acceptance amendment v2 approved)

## 1. Decision Context

Slice 0.7.6.4.19.1 corrects the live-quality blocker recorded in docs/radar/pipelines/validation/0.7.6.4.19/BLOCKING_RCA.md. The failed live run signal-run-55a9cfb4-b0f1-48ee-8aef-df20a236266f completed all required lanes, but reproduced only two of four positive controls, reached only one negative control, and accepted one XLSX source classified as identity_only.

The source candidate run remains radar-run-b03fac86-7307-448f-8deb-c1ea1794956c. Candidate discovery, scheduling UI and candidate-universe expansion are out of scope.

## 2. AS IS Problem

The current planner contains special S1, S2 and 2026 query branches instead of taking search vocabulary from the Radar criterion. Parsed evidence is projected only against the criterion of the task that retrieved it. A valid S2 source returned by an S1 task can therefore be lost. Capability validation rejects a response only when every returned source is identity-only, allowing an identity XLSX to survive inside a mixed response. Finally, all previous signal outputs for a Radar are loaded into every new run, so two genuinely independent initial runs cannot be proven without deleting history.

## 3. Intended Flow

Initial A and B use different monitoring_series_id values and therefore both start with empty fingerprints, source keys and watermarks. Incremental C uses the B series and loads only B history. The normal default series preserves existing product behavior.

## 4. Criterion-Owned Planning

Each signal criterion exposes additive search_terms and evidence_match_terms. The planner combines candidate legal name, aliases, criterion terms, source lane and effective window. It does not branch on criterion codes or a calendar year. Each candidate/criterion pair receives a stable criterion_obligation_id, a primary query and at most one alternate query. Search terms are product configuration, not benchmark controls.

The SIBUR benchmark definition contains generic TOIR/reliability vocabulary for S1 and modernization/capacity vocabulary for S2. It contains no control URLs, event dates or expected benchmark answers.

## 5. Cross-Criterion Evidence Reconciliation

Successful parsed provider output remains available in memory as a product-safe evidence bundle. After retrieval and before the final checkpoint, evidence is tested against every other enabled criterion for the same candidate.

Reuse is accepted only when entity matching, configured criterion phrases, source binding, source capability and temporal validation all pass. The target observation receives its own validation decision and keeps the origin task, origin criterion, receipt and source ref. Reuse never marks another lane as executed and never advances its watermark. Evidence from another candidate, without configured criterion terms, or matching only a broad keyword is retained as a rejected reconciliation record.

## 6. Capability And URL Semantics

Capability is checked per evidence ref. identity_only and registry refs can remain as identity or review provenance but cannot be included in confirmed source refs and always have score zero. Structured XLS/XLSX/CSV/XML/JSON files cannot be promoted from product-safe text alone. Mixed responses confirm only through independently signal-capable refs.

A shared canonical URL identity is used by evidence validation, dedupe and the post-run evaluator. Fragments and tracking parameters (utm_*, erid, gclid, yclid) are ignored. Host, path and non-tracking query parameters remain strict; domain-only control matching is forbidden.

## 7. Acceptance Freeze And Audit

Before live execution, the adjacent acceptance manifest is frozen into a session file containing its SHA-256 and git commit. The acceptance runner checks the hash before A, B and C. Controls are never included in input snapshots, tasks, queries, source contracts or provider requests. If the frozen manifest changes, the run fails. The original v1 manifest and its machine FAIL remain archived byte-for-byte. After the bounded five-cycle RCA, an explicitly reviewed v2 amendment may change only the reproducibility criterion; controls, accepted URLs, dates and semantic integrity requirements remain unchanged.

The machine report contains a per-control cross-run matrix with states found_in_both, found_only_in_a, found_only_in_b, found_in_neither, found_under_other_criterion and found_and_rejected. Aggregate observed count cannot close the slice.

## 8. Requirements And Tests

- SM-REP-01: one byte-identical frozen manifest is used for A, B and C.
- SM-REP-02: A and B independently start with empty history.
- SM-REP-03: each initial run finds at least three positive controls, one run finds all four, and the union of both runs finds all four.
- SM-DRIFT-01: the only accepted per-run miss is explicitly classified as provider search drift and cannot hide a semantic, temporal or source-policy failure.
- SM-QUERY-02: query obligations come from criterion configuration.
- SM-XCRIT-01: valid cross-criterion evidence is independently validated.
- SM-XCRIT-02: irrelevant or cross-candidate evidence is never copied.
- SM-CAP-03: identity-only and registry refs confirm zero fresh signals.
- SM-URL-01: URL identity ignores tracking only and remains host/path strict.
- SM-DED-03: C inherits B and republishes no previous signal or review item.
- SM-PROC-03: RCA, TO BE, manifest, tests, three live reports, validation and finalized AS IS are traceable.

## 9. Hard Acceptance

Both A and B must contain exactly six evidence-complete candidates, three accepted and three review-needed, two criteria and twelve pairs. Each must find at least three of four positive controls, one initial run must find all four, and their aggregate must find all four. A per-run miss is accepted only for the explicitly frozen khimprom-modernization-automation-2025 control and is classified as provider_search_drift. Each run must reject at least two frozen negative controls, retain at least one unknown-date control and contain zero false-positive, identity-confirmed, receipt-gap, orphan-decision, false-not-observed, score-zero-confirmed or required-budget-limited records.

C must use per-lane B watermarks, load prior source keys, republish zero old confirmed or review items and advance only successful lanes. All three reports must remain byte-stable and readable after API/worker restart. Only a machine-generated PASS, this document marked Implemented, and reconciled AS IS documentation allow the slice to become Done.

## 10. Open Questions

The exact-URL stability issue is intentionally deferred to dedicated provider search-routing experiments. A failed aggregate control, a run below three of four, or any semantic integrity failure remains an acceptance failure.

## 11. Approved Acceptance Amendment

The original v1 manifest SHA-256 9dfab1ee6a2a449109d35b8cf53b097cae3a4b48797bfedfb4c7214df2d6d82e and its FAIL report are preserved as *.v1.json / VALIDATION_REPORT_V1.md.

The approved amendment follows the user decision recorded after the bounded autofix limit was exhausted. It does not claim that the second run passed the old DoD and does not change any benchmark control. It separates functional pipeline correctness from provider-search stability, which is tracked as follow-up work.